

Roll No.

TAS-501

**B. TECH. (MECHANICAL ENGINEERING)
(FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

AERODYNAMICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

(iii) Draw neat diagrams wherever necessary.

(iv) Make assumptions for any missing data.

1. (a) With a neat sketch, show the primary control surfaces of the aircraft and explain the moment produced by each. (CO1)

OR

- (b) Define the four forces acting on the flight lift, weight, thrust and drag. Also draw a neat sketch. (CO1)

2. (a) What do you mean by a dimensionless number ? What is its significance in flow analysis ? (CO1)

OR

- (b) Define and bring out the significance of the following non-dimensional number : (CO1)

- (i) Reynolds' number

P. T. O.

(2)

- (ii) Mach's number
- (iii) Weber's number
- (iv) Froude's number

3. (a) Write the contribution of aircraft evolution towards mankind. (CO2)

OR

(b) Explain in detail the design configuration of the first heavier than aircraft developed by the Wright brothers. (CO2)

4. (a) List the types of drag associated with viscous fluid and explain each in detail. (CO3)

OR

(b) Explain. What do you understand by aerodynamic heating in viscous flow? CO3

5. (a) With the help of neat labelled diagrams, explain the airfoil-flap-slat configuration during : (CO3)

- (i) takeoff
- (ii) landing.

OR

(b) Draw an unsymmetrical airfoil, explain the following terminology : (CO3)

- (i) Camber line
- (ii) Angle of attack
- (iii) Taper ratio
- (iv) Wing loading
- (v) Swept wing
- (vi) Swept angle